CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (CE) Examination

November 2013

CE303 Computer Graphics

Date: 27.11.2013, Wednesday **Time: 10:00 a.m. To 01:00 p.m.**
Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) **Answer the following questions.** [06]
1. What is Computer Graphics? List out the applications of it.
2. Explain DVST.

(b) Explain Raster Scan System in detail. [05]

Q - 2 (a) Explain Beam penetration method and Shadow mask method in detail. [04]

(b) Explain Boundary Fill algorithm and differentiate 4-Connected and 8-Connected methods. [04]

(c) Derive Bresenham's Line Drawing algorithm for $|m| < 1$ and write down the equations for $|m| > 1$. [04]

OR

Q - 2 (a) Consider the line from (15, 10) and (20, 22). Use the Bresenham's Line Drawing algorithm for rasterizing the line. [03]

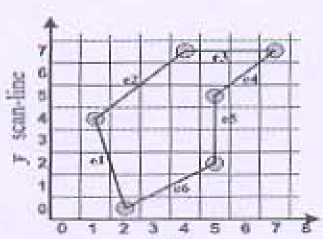
(b) **Answer the following questions.** [05]
1. Differentiate Raster Scan Displays and Random Scan Displays.
2. Differentiate Boundary-Fill algorithm and Flood-Fill algorithm.

(c) Explain Scan Line Polygon Fill algorithm. What is the disadvantage of the algorithm? Write down the method to solve the disadvantage. [04]

Q - 3 (a) Explain Refresh Cathode-Ray Tube with all components in detail. [04]

(b) Derive Midpoint circle drawing algorithm for the octant where slope of the curve varies from 0 to –1. [04]

(c) For a given polygon, prepare the Edge Table (ET) and Active Edge Table (AET) used in scan-fill polygon filling algorithm. [04]



OR

(c) Explain odd-even rule to identify interior region of self-intersecting edges of polygon. [04]

SECTION – II

Q - 4 (a) Do as Directed. **[07]**

1. Prove that the multiplication of 2D transformation matrices for each of the following sequence of operations is commutative. [03]
   i. Two successive rotations.
   ii. Two successive translations.
2. A line end points are A (0, 6) and B (6, 0). Reflect an object having vertices P (5, 3), Q (5, 5), R (3, 5), S (3, 3) with respect to given line. [04]

(b) What is the problem with Sutherland-Hodgeman polygon clipping? Explain Weiler-Atherton Polygon clipping algorithm with example. **[04]**

Q - 5 (a) How can we rotate an object with respect to any arbitrary line in 3-D? Derive matrices. **[05]**

(b) Do as Directed. **[07]**

1. Two end-points of a line: A (20, 65) and B (80, 30). Rectangle Clip window is (30, 20) and (75, 50). Clip the line using Liang-Barsky Line Clipping algorithm. [04]
2. What are the disadvantages of Cohen-Sutherland Line Clipping algorithm and compare it with Nicholl-Lee-Nicholl Line Clipping algorithm. [03]

OR

Q - 5 (a) What is 2D shear transformation? Apply shear transformation for $sh_y = ½$ relative to $x_{ref} = -1$ on the square A (0, 0), B (1, 0), C (0, 1), D (1, 1). Draw initial and final shape with coordinates. **[04]**

(b) Answer the following questions. **[08]**

1. Explain the term region codes. Write the steps of Cohen-Sutherland line clipping algorithm. [05]
2. Translate the triangle A(0,0,0), B(2,4,4), and C(10,10,4) with T = (3,5,9) using homogeneous coordinates-four elements column matrices. [03]

Q - 6 (a) Explain Back-Face Detection algorithm. **[06]**

(b) Explain Depth-Buffer method. **[06]**

OR

Q - 6 (a) Write short note on RGB Model. **[06]**

(b) Explain Nicholl-Lee-Nicholl Line Clipping algorithm with all cases. **[06]**

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